Beyond thermal accumulation: improving cross-regional grapevine phenology prediction with terrain-aware forcing and residual learning

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Abstract

Classical temperature-driven process models reduce phenological development to thermal accumulation toward fixed stage thresholds, assuming that temperature exposure and thermal responses remain sufficiently stable across sites and developmental stages. This simplification can generate spatially structured errors where terrain distorts gridded temperature estimates and later phenological stages respond to additional biophysical and management controls. To address these limitations, stage-specific process models were combined with terrain-adjusted temperature forcing and machine-learning residual correction. The framework was evaluated under nested spatially blocked validation using 23,170 TEMPO and DWD phenological records from 707 sites and 12,209 site-years during 1951–2026; machine-learning models were trained exclusively on spatially out-of-fold process-model residuals. The selected process models produced RMSE values of 9.6, 8.2, 15.5 and 30.5 d for budburst (BBCH07), flowering (BBCH65), veraison (BBCH81) and maturity/harvest (BBCH89), respectively. Terrain adjustment of ERA5-Land temperature produced limited changes for budburst and flowering but reduced RMSE by 1.3 and 2.2 d for veraison and maturity/harvest. Residual correction further reduced RMSE to

8.3, 7.7, 14.6 and 10.6 d, corresponding to improvements of 12.7%, 6.0%, 5.8% and 65.4%, respectively. MAE and RMSE decreased in all five held-out spatial test sets, and the pronounced late-season bias of the process-only maturity/harvest model was nearly eliminated. Ablation and SHAP analyses associated the corrections with process-model timing, temperature structure, terrain and spatial variability, atmospheric water demand and radiation. Thermal-time models therefore provide an interpretable biophysical foundation, while terrain-aware forcing and residual learning address complementary errors in environmental inputs and process representation. Their integration improves the accuracy and spatial transferability of grapevine phenology prediction, with the greatest benefit where systematic late-season errors remain.

Keywords: Grapevine, Phenology, Terrain temperature correction, Process-based models, Machine learning, Cross-regional validation

1 Introduction

Grapevine production is a globally important, high-value horticultural sector (Van Leeuwen et al., 2024). Phenological stages mark key developmental windows and are therefore central to vineyard management and cross-regional coordination. Budburst, flowering, veraison, and maturity or harvest represent four major phenological stages. Shifts in their timing can alter the overlap between sensitive developmental periods and frost, heat, water deficit, pest and disease pressure, and vineyard operations, thereby affecting grape yield and fruit composition (Jones and Davis, 2000; Tomasi et al., 2011; García de Cortázar-Atauri et al., 2017). Reliable and transferable phenology prediction is therefore important for seasonal climate-risk assessment, cross-regional vineyard management, planning for production-area shifts, and cultivar suitability evaluation.

Most operational grapevine phenology models represent development as a temperature-driven clock. Growing degree-day and GFV models accumulate forcing until a fixed thermal requirement is reached (García de Cortázar-Atauri et al., 2009; Parker et al., 2011). Budburst models additionally represent the transition from endodormancy to ecodormancy by coupling winter chilling with spring forcing, either sequentially, in parallel or with partial overlap; greater chilling generally reduces the subsequent heat requirement (Chuine, 2000). These formulations are parsimonious, but chilling units and thermal thresholds are proxies rather than direct measures of physiological state. Dormancy also involves transport, phytohormones, genetic and epigenetic regulation, and carbohydrate dynamics (Fadón et al., 2020). Model structure therefore matters: seven of twelve evaluated chilling models produced a physiologically inconsistent chilling–heat relationship and biased warming projections (Wang et al., 2020).

Recent studies have sought to relax these assumptions. PhenoFlex flexibly links Dynamic Model chilling with growing-degree-hour forcing, although warm-winter tests, calibration-data diversity and cross-regional validation still reveal transferability constraints (Luedeling et al., 2021; Fernandez et al., 2022; Picornell et al., 2025; Mojahid et al., 2025). At larger spatial scales, environmental clustering reduced heterogeneity in calibrated thermal requirements and improved twelve process models (Tan et al., 2024); non-constant formulations reduced systematic bias in nine maize models (Kang et al., 2024); and coupling legacy effects with temperature response showed that development depends partly on preceding conditions (Wu et al., 2024). Physics-guided continuous learning offers a complementary route by retaining process structure while updating systematic errors from new observations (Ou et al., 2025). These advances indicate that a biophysical baseline remains useful, but fixed responses cannot fully represent spatial adaptation, environmental memory and omitted controls.

Three related problems remain for cross-regional grapevine prediction. First, budburst, flowering, veraison and maturity/harvest differ in physiological control and observational meaning, so a model calibrated for one stage or region may not transfer to another (Lorenz et al., 1995; Ortega-Farias and Riveros-Burgos, 2019). Thermal requirements also vary with cultivar, climate, topography and developmental interval (Keller and Tarara, 2010; Cameron et al., 2022; Martínez-Lüscher et al., 2016). Stage-specific structures must therefore be tested in spatially unseen regions.

Second, gridded meteorological forcing may not represent vineyard conditions where grid-cell and site elevations differ. Terrain modifies temperature, radiation and cold-air drainage, especially in complex viticultural landscapes (Alikadic et al., 2019; Fraga et al., 2016; Verdugo-Vásquez et al., 2020). Elevation-aware temperature adjustment is physically motivated, but its predictive value requires spatially independent evaluation.

Third, process-model residuals can retain systematic effects of omitted physiology, temperature structure, water status, radiation, terrain, management and observation (Feigl et al., 2022).

Machine learning can model these residuals without replacing the process baseline (Reichstein et al., 2019; Shahhosseini et al., 2021), but random validation can overstate gains in spatially structured data. Residual models must therefore be trained on spatially out-of-fold predictions and tested in held-out regions (Roberts et al., 2017; Valavi et al., 2019).

This study evaluates whether stage-specific process models, terrain-adjusted temperature forcing and machine-learning residual correction improve the accuracy and spatial transferability of grapevine phenology prediction. Raw and adjusted temperature branches are compared within nested spatial validation, residual learners are trained only on out-of-fold process-model errors, and the resulting gains are examined through temperature-input sensitivity and mechanism-group ablation.

2 Materials and methods

2.1 Study framework

The overall study framework is shown in Fig. 1. Phenological observations, raw ERA5-Land meteorological forcing, DEM-derived terrain information and the corresponding terrain-corrected temperature series were first organized as model inputs. Candidate process models were evaluated separately for the four main BBCH phenological stages. Within each outer spatial split, calibration, residual learning and model selection were restricted to the spatial training blocks, and inner spatial out-of-fold process-model residuals were used to define the residual-learning task for candidate machine-learning models. The held-out outer spatial blocks were used only for final evaluation, producing stage-specific selected process models, hybrid predictions and gains, spatial validation performance, and residual-interpretation outputs.

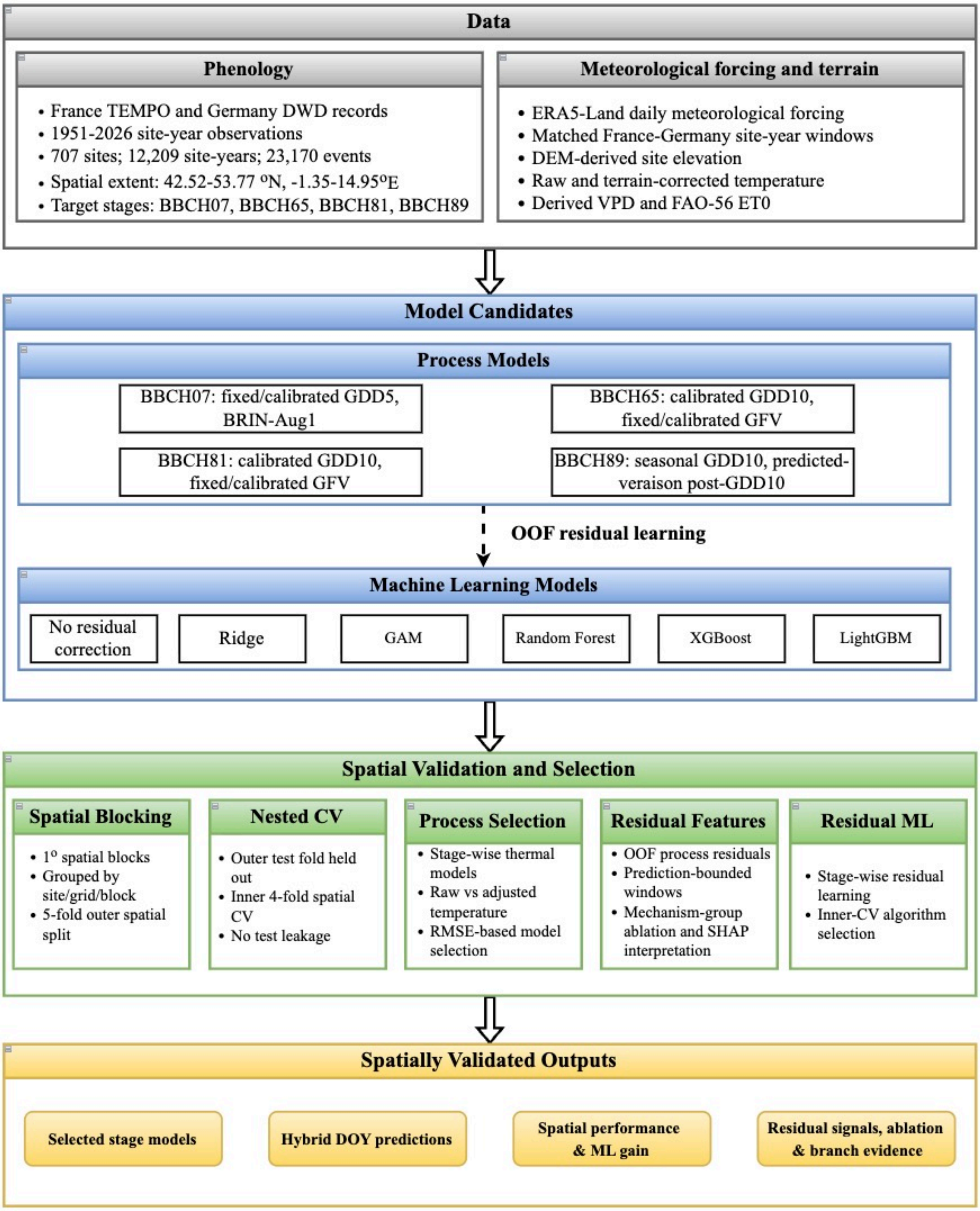
Data
Phenology
• France TEMPO and Germany DWD records
• 1951-2026 site-year observations
• 707 sites; 12,209 site-years; 23,170 events
• Spatial extent: 42.52-53.77 °N, -1.35-14.95°E
• Target stages: BBCH07, BBCH65, BBCH81, BBCH89
Meteorological forcing and terrain
• ERA5-Land daily meteorological forcing
• Matched France-Germany site-year windows
• DEM-derived site elevation
• Raw and terrain-corrected temperature
• Derived VPD and FAO-56 ET0
Model Candidates
Process Models
BBCH07: fixed/calibrated GDD5, BRIN-Aug1
BBCH65: calibrated GDD10, fixed/calibrated GFV
BBCH81: calibrated GDD10, fixed/calibrated GFV
BBCH89: seasonal GDD10, predicted-veraison post-GDD10
OOF residual learning
Machine Learning Models
No residual correction
Ridge
GAM
Random Forest
XGBoost
LightGBM
Spatial Validation and Selection
Spatial Blocking
• 1° spatial blocks
• Grouped by site/grid/block
• 5-fold outer spatial split
Nested CV
• Outer test fold held out
• Inner 4-fold spatial CV
• No test leakage
Process Selection
• Stage-wise thermal models
• Raw vs adjusted temperature
• RMSE-based model selection
Residual Features
• OOF process residuals
• Prediction-bounded windows
• Mechanism-group ablation and SHAP interpretation
Residual ML
• Stage-wise residual learning
• Inner-CV algorithm selection
Spatially Validated Outputs
Selected stage models
Hybrid DOY predictions
Spatial performance & ML gain
Residual signals, ablation & branch evidence

Fig. 1: *Workflow of this study. OOF, out-of-fold. CV, cross-validation. VPD, vapour pressure deficit. ET0, FAO-56 reference evapotranspiration.*

2.2 Study area and datasets

2.2.1 Source datasets and study area

The study used two public grapevine phenology datasets from France and Germany. French records were downloaded from the TEMPO phenological database public API (https://data.pheno.fr/api/). German records were downloaded from the Deutscher Wetterdienst (DWD, German Weather Service) open-data phenology archive (https://opendata.dwd.de/climate_environment/CDC/observations_germany/phenology/). Both source datasets include event dates, observation years, site identifiers and geographic coordinates. TEMPO records provide explicit BBCH codes, which were directly normalized to two-digit BBCH stages. DWD records use DWD stage names and identifiers, but the selected grapevine phases have the same phenological meaning as the target BBCH stages. These stages were therefore mapped to the corresponding extended BBCH stages following the grapevine BBCH standard (Lorenz et al., 1995). This harmonization provided a common stage framework for cross-source analysis. The site inventory used for data screening and blocked spatial validation contained 1,624 French sites and 700 German sites distributed across lowland, valley and upland terrain (Fig. 2). Site elevation and local terrain information were derived from the Copernicus GLO-30 digital elevation model (DEM; https://registry.opendata.aws/copernicus-dem/).

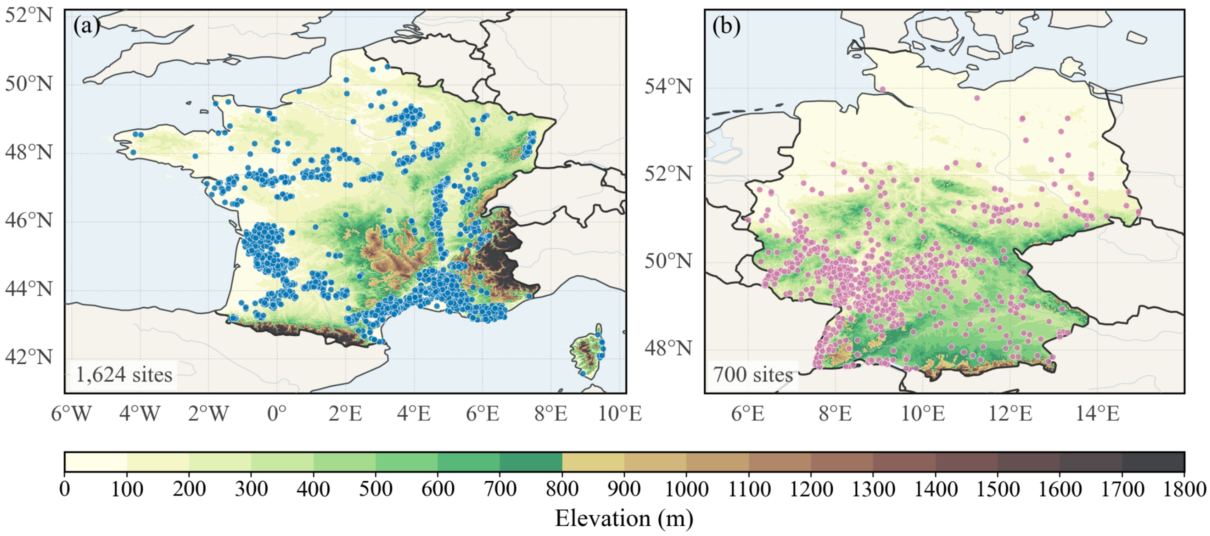
(a)
52°N
50°N
48°N
46°N
44°N
42°N
1,624 sites
6°W
4°W
2°W
0°
2°E
4°E
6°E
8°E
10°E
(b)
54°N
52°N
50°N
48°N
700 sites
6°E
8°E
10°E
12°E
14°E
0
100
200
300
400
500
600
700
800
900
1000
1100
1200
1300
1400
1500
1600
1700
1800
Elevation (m)

Fig. 2: *Spatial distribution of grapevine phenology observation sites in France and Germany.*

2.2.2 Event aggregation and main analysis sample

The event-level analysis unit was a site–year–stage record. The target stages were BBCH07 (budburst), BBCH65 (flowering), BBCH81 (veraison), and BBCH89 (maturity/harvest). Multiple source records mapped to the same site, year and target stage were aggregated to a single event using the standardized aggregation rule stored in the processed event table. Each retained event was represented by its observed day of year (DOY), site identifier, year, geographic coordinates, source dataset and target BBCH stage. Records were retained when they had valid coordinates, a target-stage DOY, DEM-derived site terrain information and a unique ERA5-Land point-weather match for the corresponding site-year. The process-model foundation dataset contained 23,671 events from 707 sites and 12,625 site-years. After applying the final residual-learning eligibility filter, which required a valid process-model prediction and complete retained residual-learning predictors, the main analysis sample contained 23,170 events from 707 sites and 12,209 site-years, covering 1951–2026 and spanning 42.52–53.77°N and -1.35–14.95°E. Stage-specific event counts in this final main-analysis sample are reported in Table 1.

Table 1: *Stage-specific event counts in the final main-analysis sample after residual-learning eligibility filtering.*

Target stage	Phenological event	Events (n)
BBCH07	Budburst	3,646
BBCH65	Flowering	4,288
BBCH81	Veraison	4,325
BBCH89	Maturity/harvest	10,911
Total		23,170

2.3 Meteorological forcing and terrain-based temperature correction

Daily meteorological forcing was obtained through Google Earth Engine from the ERA5-Land Daily Aggregated collection (collection ID: ECMWF/ERA5_LAND/DAILY_AGGR) for the corresponding weather period of each retained site–year (Muñoz-Sabater et al., 2021; Copernicus Climate Change Service, 2019). Because vineyard sites are distributed across heterogeneous terrain, DEM-derived site elevation was used to adjust ERA5-Land daily air-temperature series from the matched grid-cell elevation to the site elevation. Elevation-based temperature adjustment is a common method for adapting gridded climate fields to terrain-dependent local temperature conditions (Daly et al., 2008). This study used the standard-atmosphere lapse rate of 6.5 °C km^{-1} as the correction parameter (International Organization for Standardization, 1975):

$$T_{adj,d} = T_{raw,d} - \gamma\left(z_{site} - z_{grid}\right), \quad \gamma = 0.0065\ {}^{\circ}\mathrm{C\ m^{-1}} \tag{1}$$

where z_{site} and z_{grid} denote the DEM-derived site elevation and the matched ERA5-Land grid-cell elevation.

The correction was applied consistently to daily mean, minimum and maximum temperature. Non-temperature meteorological variables were taken directly from the matched ERA5-Land grid cell. Because the process models in this study were temperature-driven, raw-temperature and terrain-adjusted-temperature branches were both used as forcing inputs to quantify the effect of terrain-temperature adjustment.

2.4 Candidate process-based phenology models

Candidate process models were defined separately for the four target stages because dormancy release, reproductive development, berry ripening and maturity/harvest represent key phenological transitions with different physiological controls. All candidate structures were temperature-driven threshold models. For GDD-type cumulative forcing models, daily effective thermal forcing was calculated as:

$$GDD_d = \max\left(T_{\mathrm{mean},d} - T_b, 0\right) \tag{2}$$

The predicted event date was the first day on which accumulated forcing reached a stage-specific threshold.

For BBCH07, we compared fixed-parameter GDD5, calibrated GDD5 and the BRIN chilling–forcing model. GDD5 served as a conventional thermal-time benchmark for grapevine budburst, whereas BRIN represented an established two-step budburst framework (García de Cortázar-Atauri et al., 2009); its chilling–forcing logic is consistent with the unified budburst modelling framework of Chuine (2000). For BBCH65 and BBCH81, GDD10 was used as a standard grapevine thermal-time comparator (Zapata et al., 2017). The Grapevine Flowering Veraison (GFV) model was also included for flowering and veraison of *Vitis vinifera*, using the fixed DOY

60 start and 0 °C base temperature proposed by Parker et al. (2011). For BBCH89, GDD10 accumulation was started either on 1 January or from the fold-specific process-predicted BBCH81 date. This stage-to-stage accumulation setting follows previous cumulative degree-day applications to post-veraison berry development and harvest-related stages (Molitor et al., 2014; Zapata et al., 2017).

Table 2: *Candidate process models evaluated for the four target grapevine phenological stages.*

Stage	Candidate model	Start and thermal response	Calibrated parameters
BBCH07	Fixed GDD5	1 January, $T_b = 5$ °C	None
	Calibrated GDD5	1 January, $T_b = 5$ °C	Forcing threshold F^*
	BRIN-Aug1	Previous 1 August, chilling and forcing phases	Chilling and forcing thresholds
BBCH65	Calibrated GDD10	1 January, $T_b = 10$ °C	Forcing threshold F^*
	Fixed GFV	DOY 60, $T_b = 0$ °C	None
	Calibrated GFV	DOY 60, $T_b = 0$ °C	Forcing threshold F^*
BBCH81	Calibrated GDD10	1 January, $T_b = 10$ °C	Forcing threshold F^*
	Fixed GFV	DOY 60, $T_b = 0$ °C	None
	Calibrated GFV	DOY 60, $T_b = 0$ °C	Forcing threshold F^*
BBCH89	Seasonal GDD10	1 January, $T_b = 10$ °C	Forcing threshold F^*
	Post-veraison GDD10	Fold-specific predicted BBCH81 date, $T_b = 10$ °C	Forcing threshold F^*

2.5 Candidate machine-learning residual models

Machine learning was used as a residual-learning component rather than as a direct phenology-date model. For event i, the signed residual and the resulting hybrid prediction were defined as:

$$\begin{aligned} r_i &= DOY_{\mathrm{obs},i} - \widehat{DOY}_{\mathrm{proc},i}, \\ \widehat{DOY}_{\mathrm{hyb},i} &= \widehat{DOY}_{\mathrm{proc},i} + \hat{r}_i \end{aligned} \tag{3}$$

Positive residuals indicate observed events later than the process-model prediction, whereas negative residuals indicate earlier observed events.

The candidate machine-learning models were selected to span simple shrinkage, smooth additive response and tree-ensemble structures (Table 3). Ridge regression provided a regularized linear benchmark. The GAM implementation represented additive nonlinear responses using spline-expanded predictors with ridge penalization. Random forest, XGBoost and LightGBM represented increasingly flexible tree-ensemble models. All candidate machine-learning models were applied to the same signed-residual target and were combined with the selected process-model prediction in the same hybrid form.

Table 3: *Candidate machine-learning models used for residual learning in the hybrid framework.*

Learner	Software package	Model structure	References
Ridge	scikit-learn 1.5.1	Penalized linear regression with L_2 shrinkage	Hoerl and Kennard (1970), Pedregosa et al. (2011)
GAM	scikit-learn 1.5.1	Additive spline basis with ridge penalization	Wood (2017), Pedregosa et al. (2011)
RF	scikit-learn 1.5.1	Bagged regression-tree ensemble with random feature subspaces	Breiman (2001), Pedregosa et al. (2011)
XGBoost	xgboost 2.1.4	Regularized gradient-boosted regression trees	Chen and Guestrin (2016)
LightGBM	lightgbm 4.7.0	Gradient-boosted decision trees with histogram-based growth	Ke et al. (2017)

2.6 Residual feature selection and mechanism groups

Residual-learning feature sets were fixed separately for each target phenological stage. Candidate variables were retained according to four criteria: sufficient coverage in the final main-analysis sample, definition from process-model predicted dates rather than observed target-stage dates, physiological relevance to residual formation, and absence of severe collinearity among predictors. Collinearity was screened using pairwise absolute correlation and variance inflation factors (VIF), following common data-exploration and regression-modelling diagnostics (Zuur et al., 2010; Dormann et al., 2013; O'Brien, 2007). The screening thresholds were $|r| < 0.80$ and $VIF < 5$. The final feature profiles contained 11 predictors for BBCH07, 12 for BBCH65, 12 for BBCH81 and 11 for BBCH89.

For BBCH07, preseason chill and frost features were summarized from the previous 1 August to the end of February, consistent with the BRIN-Aug1 input period. Other pre-budburst features ended at the process-predicted BBCH07 date. For BBCH65, BBCH81 and BBCH89, interval meteorological features were summarized from the process-predicted upstream stage to the process-predicted target stage. Short pre-event windows of 15 or 30 days used the process-predicted target-stage date as the endpoint. VPD and ET0 were calculated following FAO-56 formulations, and positive climatic water deficit was defined as the positive part of ET0 minus precipitation (Allen et al., 1998).

The fixed predictors were then grouped by mechanism for feature-group ablation and feature-level interpretability analysis (Table 4). This grouping did not alter the residual target, the hybrid prediction equation or the spatial validation framework.

Table 4: *Fixed residual-learning feature profiles and mechanism groups.*

Stage	Feature group	Retained features	Summary descriptor
BBCH07	Process state	Process-predicted DOY	Process-model timing
	Temperature structure	Preseason chill days, preseason frost degree-days, pre-07 15-d GDD5 rate, pre-07 30-d GDD5 rate	Dormancy cold exposure and pre-budburst forcing rate
	Radiation	Pre-07 15-d shortwave radiation sum	Pre-budburst radiation input
	Terrain/spatial	Elevation, local elevation SD, northness, latitude, longitude	Site terrain, aspect and spatial position
BBCH65	Process state	Process-predicted DOY	Process-model timing
	Temperature structure	07–65 mean DTR, 07–65 cool degree-days, 07–65 GDD10 rate, pre-65 30-d DTR	Thermal range, cool exposure and forcing rate
	Water/drought	07–65 mean VPD, 07–65 ET0 sum, 07–65 positive CWD sum, pre-65 15-d mean VPD	Atmospheric demand and water deficit
	Radiation	07–65 mean shortwave radiation	Budburst-to-flowering radiation input
	Terrain/spatial	Elevation, northness	Site elevation and aspect
BBCH81	Process state	Process-predicted DOY	Process-model timing
	Process timing	Predicted 65–81 stage length	Predicted phase duration
	Temperature structure	65–81 Tmean SD, 65–81 mean DTR, 65–81 heat degree-days above 30 °C, pre-81 30-d heat degree-days above 35 °C, pre-81 15-d GDD10 rate	Temperature variability, diurnal range and heat exposure
	Water/drought	Pre-81 30-d positive CWD sum	Pre-veraison water deficit
	Terrain/spatial	Local elevation SD, northness, latitude, longitude	Terrain complexity, aspect and spatial position
BBCH89	Process state	Process-predicted DOY	Process-model timing
	Temperature structure	81–89 Tmean SD, pre-89 15-d GDD10 rate, pre-89 15-d heat degree-days above 30 °C	Temperature variability and late-season heat exposure
	Water/drought	81–89 precipitation sum, 81–89 mean VPD, 81–89 positive CWD sum	Ripening-period moisture and atmospheric demand
	Radiation	81–89 shortwave radiation sum	Veraison-to-maturity radiation input
	Terrain/spatial	Elevation, local elevation SD, northness	Site elevation, terrain complexity and aspect

2.7 Model training, selection and spatial validation

2.7.1 Two-level spatial blocking and data partitioning

All model calibration, model selection and testing were conducted within a two-level blocked spatial-validation framework (Fig. 3). This study constructed 1° spatial blocks from the ERA5-Land grid-centre coordinates matched to the observation sites, and used these blocks as the indivisible partitioning units. When observation sites are spatially close or share similar gridded meteorological forcing, random splitting can overestimate apparent predictive performance

(Roberts et al., 2017; Valavi et al., 2019; Schratz et al., 2019). The 1° block size was therefore used as a compromise: it prevented nearby sites with similar ERA5-Land forcing from being split between training and held-out sets, while avoiding overly large blocks that would leave too few spatial partitions for repeated model fitting and evaluation.

To ensure strict evaluation in spatially unseen areas, the spatial hierarchy was defined as follows. First, all 1° spatial blocks containing valid records were assigned to five outer spatial groups using a fixed random seed, with fold balance checked by country and phenological stage. In each outer round, one outer spatial group was held out as the outer test set, and the remaining four outer spatial groups formed the outer training set. The outer test set was used only for final model evaluation and was not used for process-model calibration, inner validation, machine-learning model training, hyperparameter tuning or model selection. Second, within each outer training set, the 1° spatial blocks were re-partitioned into four inner spatial folds. In each inner round, three inner spatial folds formed the inner training set, and the remaining inner spatial fold served as the inner validation set. The inner validation sets supported process-model tuning and selection, machine-learning model tuning and out-of-fold residual generation. After inner selection, the selected model configuration was refitted on the full outer training set and then used to predict the held-out outer test set.

The partitioning terms used in this study are summarized in Table 5.

Table 5: *Terminology used for the two-level blocked spatial-validation framework.*

Term	Definition in this study
1° spatial block	Basic spatial partitioning unit and the smallest unit that was not split across folds.
Outer spatial group	One of five first-level groups formed by assigning 1° spatial blocks with valid records.
Outer training set	Records from the four outer spatial groups not held out in a given outer round.
Outer test set	Records from the held-out outer spatial group and used only for final spatial evaluation.
Inner spatial fold	One of four second-level folds created from 1° spatial blocks within the current outer training set.
Inner training set	Records from three inner spatial folds used for model fitting within an inner round.
Inner validation set	Records from the held-out inner spatial fold used for tuning, selection and out-of-fold residual generation.

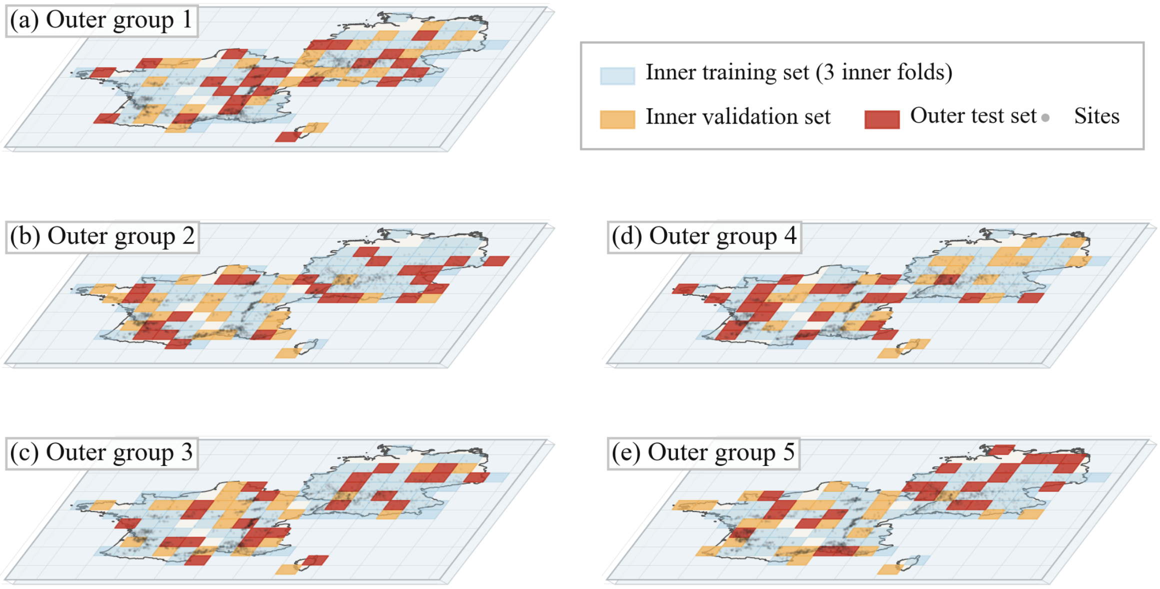

Fig. 3: *Two-level blocked spatial validation design based on 1° ERA5-Land-centred spatial blocks. Each panel shows one outer spatial group held out as the outer test set and a representative split between the inner training set and inner validation set within the corresponding outer training set.*

2.7.2 Process-model calibration and selection

Process-model calibration followed the two-level spatial partitioning framework defined above. For each phenological stage and outer round, the outer test set was excluded from all calibration and selection steps. Within the corresponding outer training set, each candidate process-model structure was evaluated separately under the raw-temperature and terrain-adjusted-temperature branches. In each inner round, model parameters were estimated only from the inner training set: fixed-parameter benchmarks were used unchanged, GDD and GFV forcing thresholds were selected by one-dimensional candidate-grid search, and BRIN chilling–forcing thresholds were selected by a two-step candidate-grid search. In all calibrated cases, the objective function was RMSE. The fitted candidate was then evaluated on the corresponding inner validation set. Model structure and temperature branch were selected using mean RMSE across the inner validation sets, with MAE used as a secondary criterion when RMSE rankings were tied or nearly tied. When the selected model structure was stable across outer rounds but the preferred temperature branch differed, the branch selected for that outer round was retained. After inner selection, the selected process model was refitted on the full outer training set and then used to predict the held-out outer test set for strict spatially held-out evaluation.

2.7.3 Residual learning and hybrid-model selection

Residual learning used the two-level spatial partitioning framework described above. It also used the selected process-model configuration from Section 2.7.2. In each outer round, the selected process model was fitted and evaluated across the four inner rounds, producing out-of-fold process-model predictions for the inner validation sets. Residuals were calculated from Eq. 3 as the difference between observed DOY and the corresponding out-of-fold process-model prediction. These residuals formed the machine-learning target within the current outer training set. Candidate machine-learning models used the same inner spatial folds for residual fitting and tuning, and predicted residual corrections for the inner validation sets.

The hybrid prediction was obtained by adding the machine-learning residual correction to the process-model prediction:

$$\widehat{DOY}_{\mathrm{hyb},i} = \widehat{DOY}_{\mathrm{proc},i} + \hat{r}_{\mathrm{ML},i} \tag{4}$$

Candidate hybrid systems, including the no-correction process-only baseline, were compared using hybrid DOY prediction errors on the inner validation sets. The machine-learning model used for each phenological stage was selected by the lowest stage-level mean RMSE, with MAE, absolute bias and model simplicity used only as tie-breaking criteria.

For final spatial evaluation, the selected ML model was fitted using residual records from the full outer training set and then applied to the held-out outer test set. The resulting outer-test predictions were used to evaluate whether residual correction improved generalization to spatially unseen records.

2.8 Performance metrics and statistical summaries

The following metrics were used to summarize prediction errors, paired model differences and fold-level variation. For evaluated record i, y_i denotes the observed phenological day of year,

$\hat{y}_i$ denotes the predicted day of year, and n is the number of evaluated records.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |\hat{y}_i - y_i| \tag{5}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (\hat{y}_i - y_i)^2} \tag{6}$$

$$Bias = \frac{1}{n}\sum_{i=1}^{n} (\hat{y}_i - y_i) \tag{7}$$

MAE, RMSE and Bias summarize absolute, squared and signed prediction errors, respectively. Positive Bias indicates later predictions than observations, and negative Bias indicates earlier predictions.

For any metric M, paired model differences were defined as:

$$\Delta M = M_A - M_B \tag{8}$$

where A denotes the tested model or branch and B denotes the reference model or branch. Negative ΔM values indicate lower error for A. The relative gain was calculated as:

$$Gain_M = \frac{M_B - M_A}{M_B} \times 100\% \tag{9}$$

For fold-specific metric M_k, the fold mean and standard error were:

$$\overline{M} = \frac{1}{K}\sum_{k=1}^{K} M_k, \qquad SE = \frac{sd(M_k)}{\sqrt{K}}, \qquad K = 5 \tag{10}$$

Worst-fold performance was defined as $\max_k M_k$ for fold-specific MAE or RMSE.

3 Results

3.1 Spatial pattern and magnitude of terrain-temperature correction

Terrain-based temperature correction showed a clear spatial pattern across the French and German phenology sites (Fig. 4). Correction magnitudes were low to moderate over most of the study region, whereas larger corrections occurred where vineyards were located in more topographically complex areas. At the country scale, Fig. 4 further shows that low-magnitude corrections accounted for a larger proportion of the French records, while medium- to high-magnitude corrections accounted for a larger proportion of the German records.

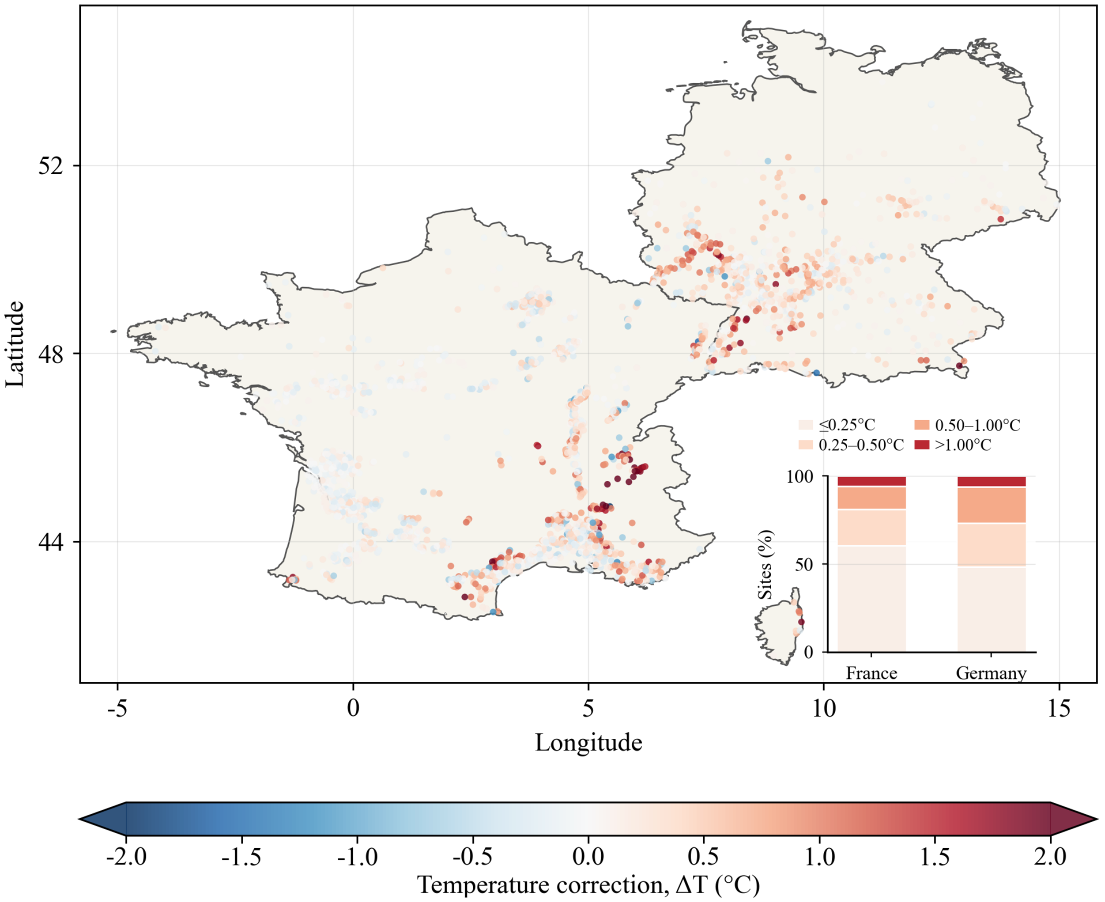
52
48
44
Latitude
-5
0
5
10
15
Longitude
≤0.25°C
0.25–0.50°C
0.50–1.00°C
>1.00°C
100
50
0
Sites (%)
France
Germany
-2.0
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
2.0
Temperature correction, ΔT (°C)

Fig. 4: *Spatial distribution and magnitude of terrain-based temperature correction across the French and German grapevine phenology sites. The inset summarizes the country-level proportional distribution of correction magnitudes.*

3.2 Process-model performance and terrain-temperature effects

Process-model performance varied markedly among phenological stages and model structures (Fig. 5). Based on RMSE, the best-performing process models for BBCH07, BBCH65, BBCH81 and BBCH89 were BRIN, GFV, GFV and the stage-to-stage GDD10 model, respectively. For the temperature-forcing branch, BBCH07 retained the branch selected within each outer round, whereas BBCH65, BBCH81 and BBCH89 selected the terrain-adjusted temperature branch.

The selected process-model baselines reached MAE/RMSE values of 6.8/9.6 d for BBCH07, 5.7/8.2 d for BBCH65, 12.4/15.5 d for BBCH81, and 27.8/30.5 d for BBCH89 (Fig. 5a,b). Flowering showed the highest prediction accuracy, whereas maturity/harvest had markedly larger errors. Observed–predicted comparisons further showed that BBCH65 strict spatial outer-fold predictions were densely distributed near the 1:1 reference line, with small model bias. BBCH07 and BBCH81 showed broader point dispersion, whereas BBCH89 exhibited a pronounced overall negative bias of -27.0 d (Fig. 6).

Terrain-adjusted temperature produced only small changes in the selected budburst and flowering models, but larger improvements for later phenological stages (Fig. 5c,d). The RMSE change was -0.06 d for BBCH07 and -0.14 d for BBCH65, compared with -1.28 d for BBCH81 and -2.17 d for BBCH89. Within the same phenological stage, the magnitude of the temperature-adjustment effect also varied among process-model structures.

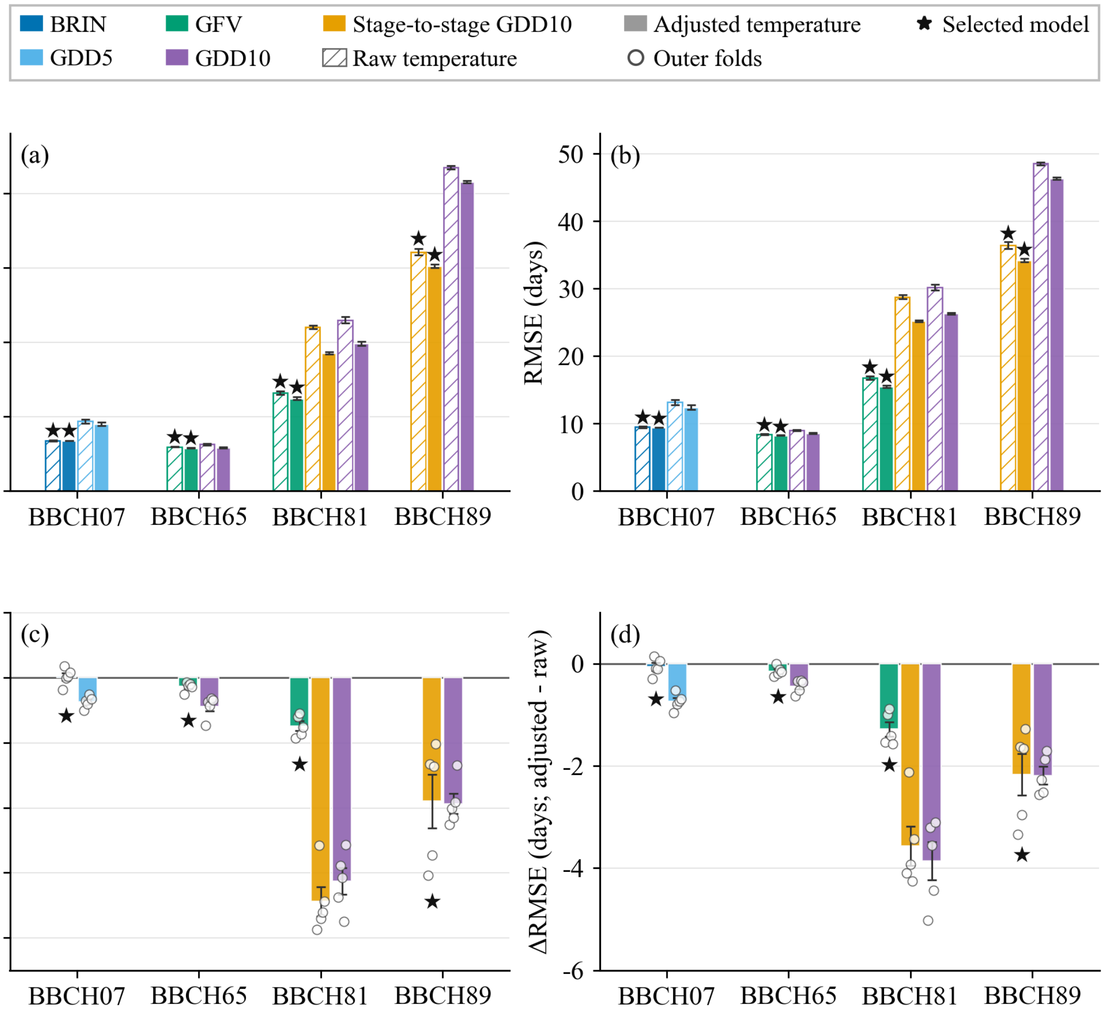

Fig. 5: *Simulation performance of candidate process models driven by raw and terrain-adjusted temperatures (a–b). Bars show means across outer rounds, and open circles show individual outer-test estimates. Panels (c–d) show error differences between terrain-adjusted and raw temperature inputs. Negative values indicate reduced process-model prediction error after terrain adjustment.*

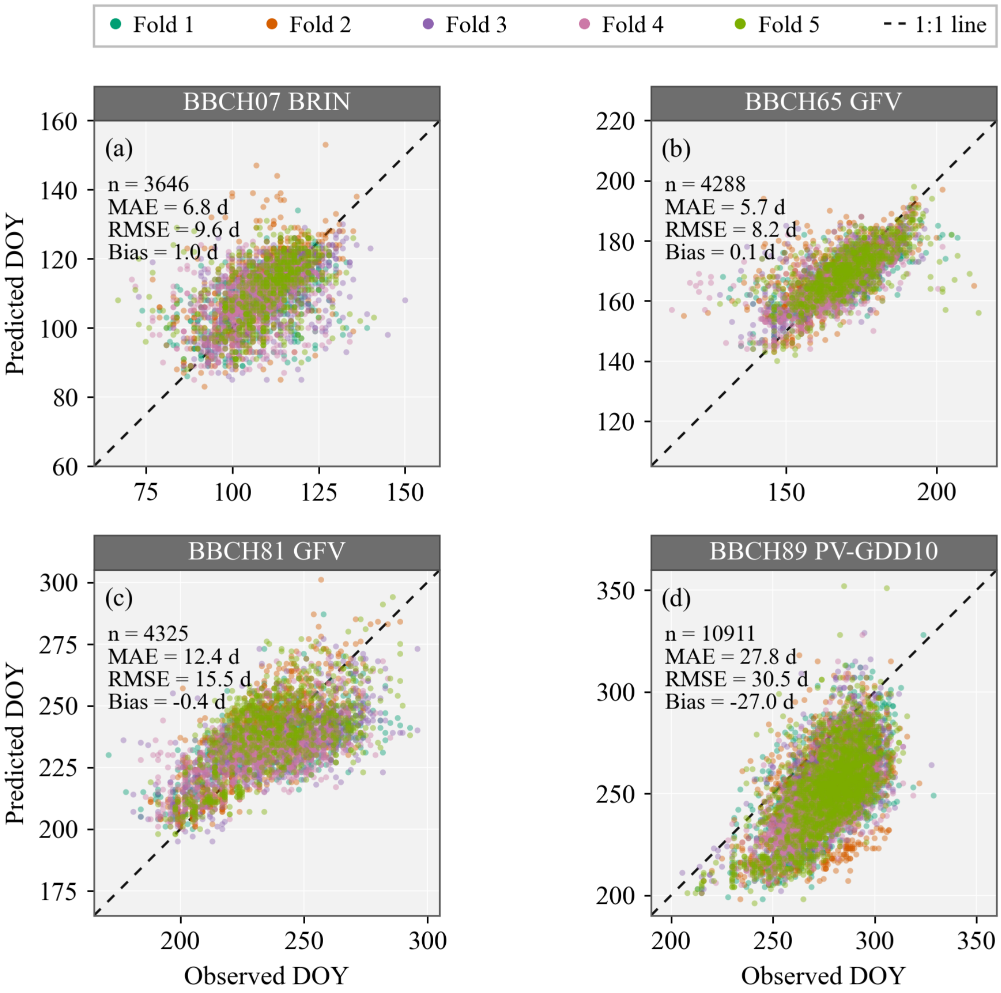

Fig. 6: *Observed and predicted phenological day of year for the selected process-model baseline at each stage. Points are outer-test predictions, and colors denote outer spatial groups.*

3.3 Machine-learning model selection and hybrid-model accuracy gains

Hybrid models with machine-learning residual correction improved upon the selected process-model baseline in all four phenological stages (Fig. 7). Based on RMSE, the fixed hybrid combinations were Ridge for BBCH07, RF for BBCH65, Ridge for BBCH81, and RF for BBCH89 (Fig. 7a,b). From BBCH07 to BBCH81, candidate machine-learning models showed similar and modest improvements, whereas the improvement was much larger for BBCH89.

RMSE decreased from 9.6 to 8.3 d for BBCH07, from 8.2 to 7.7 d for BBCH65, from 15.5 to 14.6 d for BBCH81, and from 30.5 to 10.6 d for BBCH89. These reductions corresponded to RMSE decreases of 12.7%, 6.0%, 5.8%, and 65.4%, respectively. MAE showed the same direction of change as RMSE, with the largest absolute reduction again observed for BBCH89 (Fig. 7c,d).

Observed–predicted comparisons of the fixed hybrid models showed near-zero aggregate bias in all four phenological stages (Fig. 8). BBCH65 remained densely clustered around the 1:1 reference line. BBCH07 and BBCH81 still showed broader point dispersion, but their aggregate biases were reduced relative to the process-only models. For BBCH89, the fixed hybrid model substantially reduced the negative bias observed in the process-only model, changing the aggregate bias from -27.0 to -0.1 d.

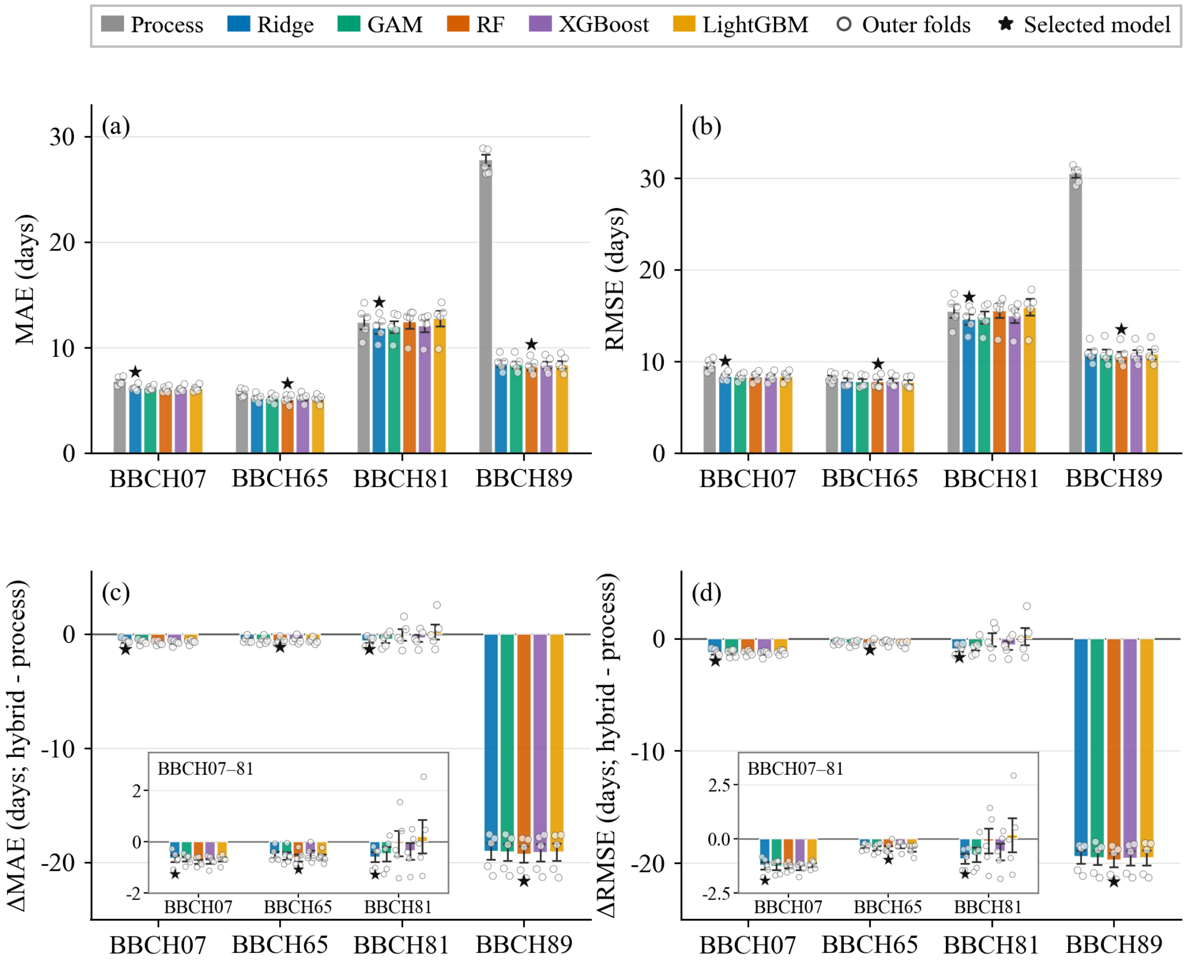

Fig. 7: *Residual-learning performance of machine-learning models for the main phenological stages (a–b). Open circles show individual outer-test estimates from the five outer rounds. Error changes of hybrid models relative to the process-only models are shown in (c–d). Negative values indicate reduced prediction error after residual correction.*

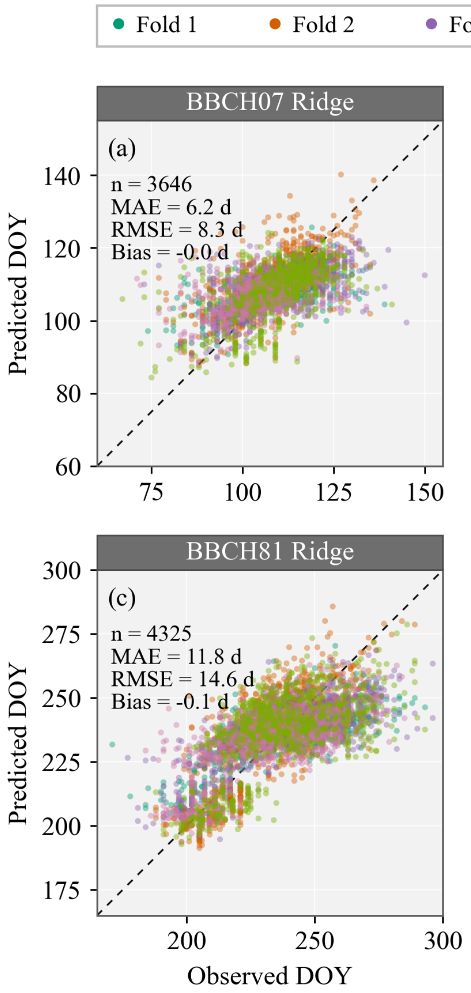
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5
1:1 line
BBCH07 Ridge
(a)
n = 3646
MAE = 6.2 d
RMSE = 8.3 d
Bias = -0.0 d
BBCH65 RF
(b)
n = 4288
MAE = 5.1 d
RMSE = 7.7 d
Bias = 0.2 d
BBCH81 Ridge
(c)
n = 4325
MAE = 11.8 d
RMSE = 14.6 d
Bias = -0.1 d
BBCH89 RF
(d)
n = 10911
MAE = 8.2 d
RMSE = 10.6 d
Bias = -0.1 d
Predicted DOY
Observed DOY
60
80
100
120
140
75
100
125
150
120
140
160
180
200
220
150
200
175
200
225
250
275
300
200
250
300
200
225
250
275
300
325
200
250
300

Fig. 8: *Observed and predicted phenological day of year for the fixed hybrid model at each stage. Points are outer-test predictions, and colors denote outer spatial groups.*

3.4 Spatial robustness of hybrid-model gains

The accuracy gains from the hybrid models remained stable across the five outer rounds (Fig. 9). MAE and RMSE decreased in every held-out outer test set for each phenological stage. Although the magnitude of improvement differed among stages, no outer round showed a reversal in which the fixed hybrid model performed worse than the corresponding process-only model.

Residual correction also improved the worst outer-round performance. The maximum outer-test RMSE across outer rounds decreased from 10.4 to 9.0 d for BBCH07, from 8.9 to 8.5 d for BBCH65, from 17.4 to 16.0 d for BBCH81, and from 31.5 to 12.4 d for BBCH89. The mean outer-round RMSE differences were -1.2, -0.5, -0.9, and -19.7 d for BBCH07, BBCH65, BBCH81, and BBCH89, respectively.

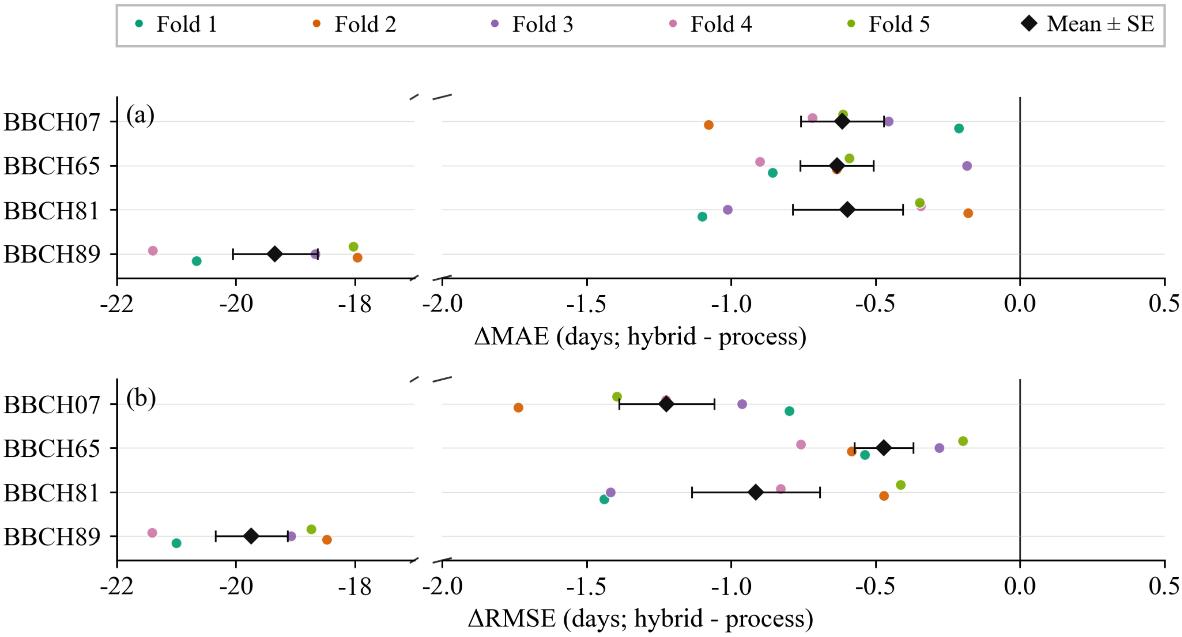

Fig. 9: *Accuracy gains of the best hybrid models across the five outer rounds. Delta values are hybrid-model error minus process-model error. Negative values indicate lower error than the selected process model. The x-axis is shown with a broken scale.*

3.5 Mechanism-group and feature-level residual interpretability

Mechanism-group ablation showed that the residual information retained by the fixed hybrid models followed a stage-dependent pattern (Fig. 10). Removing the process-state group caused the largest RMSE increase for BBCH07 (+0.87 d) and BBCH89 (+0.18 d). For BBCH65, the largest increase was associated with temperature-structure variables (+0.19 d). For BBCH81, the terrain/spatial group had the largest contribution (+0.42 d), followed by process-state and process-timing variables.

The remaining groups had smaller and less uniform effects. Water/drought variables increased RMSE when removed from BBCH65 and BBCH89, but produced little change for BBCH81. Radiation variables contributed weakly but positively in BBCH07, BBCH65 and BBCH89. Terrain/spatial removal did not increase RMSE in every stage, showing that group-level contributions were not transferable as a single common ranking across the four phenological events.

Feature-level SHAP patterns confirmed that the dominant residual predictors differed among stages (Fig. 11).

In BBCH07, process-predicted DOY was the strongest feature, followed by pre-BBCH07 GDD5 accumulation rate, latitude, elevation, shortwave radiation, frost intensity and preseason chill. In BBCH65, elevation and BBCH07–65 GDD10 accumulation rate ranked highest, followed by radiation, diurnal temperature range and northness. In BBCH81, predicted BBCH65–81 interval length, longitude and latitude dominated the SHAP ranking, with additional contributions from heat intensity, local elevation variability and pre-veraison climatic water deficit. In BBCH89, pre-BBCH89 GDD10 rate and process-predicted DOY were the leading features, followed by elevation, radiation and VPD-related variables.

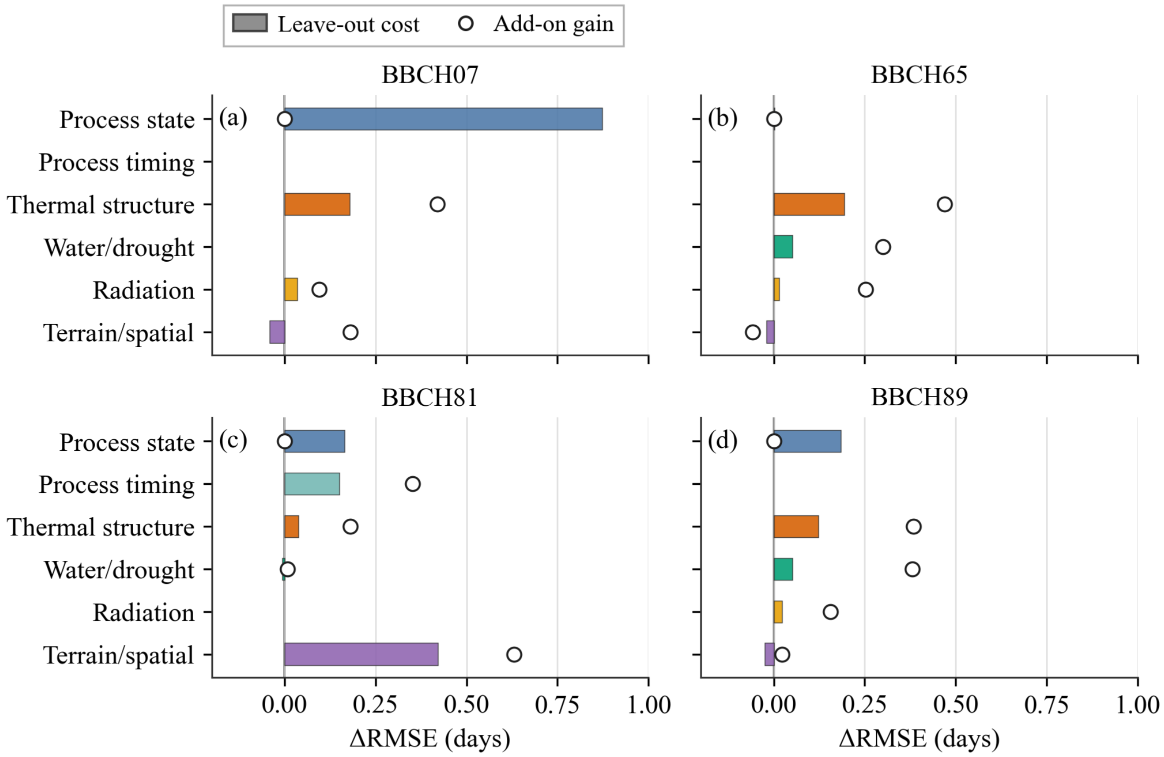

Fig. 10: *Mechanism-group ablation results for residual-learning features in the best hybrid models. Positive values indicate that removing the feature group worsens model performance.*

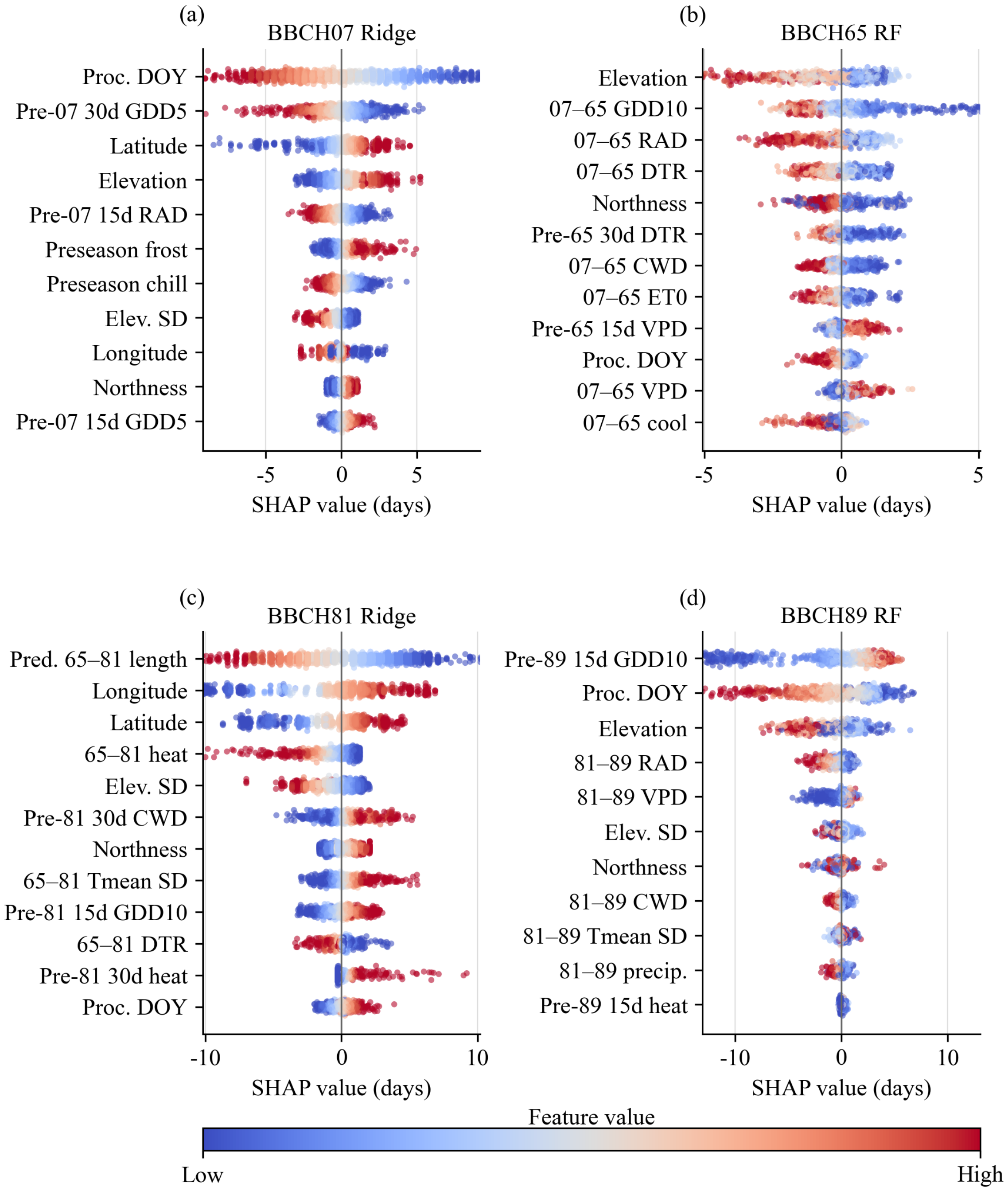

Fig. 11: *SHAP results for residual-learning features in the best hybrid models. Point color represents normalized feature values, with blue denoting low values and red denoting high values. Abbreviations: Proc. DOY, process-model predicted day of year. Pre-07/65/81/89, windows before the predicted dates of BBCH07/65/81/89. GDD, growing degree days. RAD, solar radiation. DTR, diurnal temperature range. CWD, positive climatic water deficit. VPD, vapour pressure deficit. ET0, reference evapotranspiration. SD, standard deviation.*

4 Discussion

4.1 Stage-dependent cross-regional transferability of process models

Strict validation in spatially unseen areas showed that the cross-regional transferability of process models was strongly stage-dependent. BBCH07 and BBCH65 were relatively stable, suggesting that the dominant temperature-related signals during early development were sufficiently strong to support spatially independent prediction. Budburst depends on dormancy release after chilling requirements are met and subsequent heat accumulation, whereas flowering is closely linked to accumulated heat after vegetative growth resumes (Chuine, 2000; García de Cortázar-Atauri et al., 2009; Zapata et al., 2017). In addition, these stages have comparatively clear field definitions, which reduces uncertainty in the observational target. This does not imply that early phenology is free from local controls, because frost, cultivar differences and spring temperature variability can still modify event timing (Keller and Tarara, 2010; Cameron et al., 2022).

BBCH81 showed intermediate performance, indicating that veraison remains partly governed by heat accumulation but is less fully represented by a single temperature-driven process structure. Veraison marks the onset of berry ripening and can be affected by preceding developmental duration, heat exposure, water status and source–sink relations, which may reduce model transferability across regions (Sadras and Moran, 2013; Zapata et al., 2017; Van Leeuwen et al., 2024). BBCH89 was the most difficult stage for process-only modelling, consistent with previous work showing that harvest timing is strongly affected by management-related conventions and therefore departs from a purely biological threshold (Ortega-Farias and Riveros-Burgos, 2019). This explains why the stage-to-stage GDD10 model reduced part of the late-season bias but still produced large spatially held-out errors (Molitor et al., 2014; Tomasi et al., 2011). Accordingly, cross-regional phenology prediction should be evaluated stage by stage under blocked spatial validation, rather than inferred from a single model structure or random cross-validation results (Roberts et al., 2017; Valavi et al., 2019).

4.2 Stage-Dependent Effects of Terrain-Adjusted Temperature

The magnitude of terrain-based temperature correction varied spatially, rather than representing a uniform temperature shift across the study region. This pattern is consistent with the known sensitivity of vineyard microclimate to elevation, slope exposure and local terrain configuration (Alikadic et al., 2019; Fraga et al., 2016; Verdugo-Vásquez et al., 2020). The predictive value of temperature correction followed a clear stage-dependent pattern. For BBCH07 and BBCH65, terrain adjustment produced only small changes in process-model performance, likely because both stages occur early in the annual phenological sequence and budburst in particular is not controlled by heat accumulation alone. Budburst depends on dormancy release after chilling requirements are met and subsequent forcing, whereas flowering is closely related to heat accumulation after vegetative growth resumes (Chuine, 2000; García de Cortázar-Atauri et al., 2009; Zapata et al., 2017). As a result, the effect of a first-order elevation correction may have less time to accumulate in these early stages than in later seasonal windows. In contrast, BBCH81 and BBCH89 showed larger RMSE reductions after correction. Later phenological stages integrate thermal conditions over longer seasonal windows, so small daily temperature differences can accumulate into larger differences in predicted timing (Tomasi et al., 2011; Molitor et al., 2014). This effect becomes more evident in models using longer accumulation periods or stage-to-stage late-season forcing, where terrain-related temperature mismatch can propagate through the thermal clock. The magnitude of the correction effect also differed among candidate model structures within the same stage. The contribution of this analysis is therefore not to restate that terrain affects temperature, but to quantify whether an elevation-aware correction improves phenology prediction when evaluated in unseen spatial blocks.

4.3 Spatially Stable Residual Correction and Interpretable Signals

Residual learning provided a correction layer that transferred to unseen spatial blocks. The fixed hybrid models reduced RMSE in all four phenological stages, from 9.6 to 8.3 d for BBCH07, from 8.2 to 7.7 d for BBCH65, from 15.5 to 14.6 d for BBCH81, and from 30.5 to 10.6 d for BBCH89. Gains were moderate for BBCH07–81, whereas BBCH89 showed a much larger improvement because the process-only model retained a strong systematic late-season bias. The near-removal of this directional offset indicates that the selected machine-learning model corrected a systematic component of the late-season error, rather than only reducing random dispersion. This is consistent with the fact that maturity or harvest timing combines biological ripening with ripeness criteria, management decisions and water-related constraints that are not explicitly represented in temperature-driven process models (Ortega-Farias and Riveros-Burgos, 2019; Molitor et al., 2014; Van Leeuwen et al., 2024). MAE and RMSE decreased in every held-out outer test set, and the worst outer-round RMSE also improved. Because these gains were obtained under strict spatial separation, their consistency indicates that the machine-learning residual correction captured spatially transferable error structure rather than only local or fold-specific patterns.

The similar and moderate performance of different machine-learning algorithms for BBCH07–81 is expected under this validation setting. When the process model already captures most of the dominant phenological signal, the remaining residual target becomes smaller, noisier and harder to predict. Strict spatial separation further prevents flexible machine-learning algorithms from relying on local spatial similarity, so added model complexity does not automatically translate into stronger out-of-region performance. This explains why relatively simple models such as Ridge regression remained competitive. For BBCH89, the residuals contained a larger systematic component, and the random forest correction substantially reduced the late-stage bias.

Ablation and SHAP analyses provided evidence for interpretable residual signals (Lundberg and Lee, 2017). Process-model predicted timing was important in BBCH07 and BBCH89, indicating that the baseline process prediction itself carried information about residual direction and magnitude. Temperature-structure, terrain/spatial, water-demand and radiation variables contributed in a stage-specific manner, suggesting that the residuals reflected environmental and spatial information not fully represented by the temperature-driven process models. These patterns are consistent with the view that process-model residuals can contain structured information related to missing processes, input limitations or environmental gradients (Feigl et al., 2022; Reichstein et al., 2019; Shahhosseini et al., 2021). Thus, the residual-learning component captured systematic, stage-specific environmental and spatial signals that complemented the selected process models under strict spatial extrapolation.

4.4 Limitations and Implications

Several limitations should be noted. First, cultivar information, management records and harvest criteria were not consistently available across the French and German datasets. This limitation is most relevant for BBCH89, because the observed date can reflect both biological maturity and operational harvest decisions. Second, the water-related predictors were based on proxy variables. VPD, ET0 and positive climatic water deficit describe atmospheric water demand, but they do not directly measure soil-water availability or vine water status. Third, terrain correction was applied to temperature using a fixed lapse-rate formulation. This provided a transparent first-order correction, but could not fully reproduce cold-air drainage, slope exposure, canopy environment or other vineyard-scale microclimatic processes. These limitations do not undermine the main inference because the study was designed to test regional spatial transferability using consistently available phenology, terrain and gridded-weather information. Future improvements will likely depend on more detailed cultivar and management records, direct soil or vine water-status information, and higher-resolution microclimatic forcing, which would allow the respective contributions of process-model structure, temperature representation and residual correction to be quantified more precisely.

5 Conclusions

This study evaluated three linked components of regional grapevine phenology prediction: the spatial transferability of stage-specific process models, the practical value of terrain-adjusted temperature forcing, and the accuracy and spatial stability of hybrid models based on machine-learning residual correction. Process-model performance showed a clear stage-dependent pattern: budburst and flowering were relatively stable in unseen spatial partitions, veraison showed intermediate performance, and maturity/harvest was the most difficult stage for process-only modelling. Terrain-adjusted temperature produced quantifiable, stage-dependent improvements. Effects were limited for BBCH07 and BBCH65 but larger for BBCH81 and BBCH89. Under strict spatial validation, residual learning further improved prediction accuracy for all four phenological stages and nearly removed the late-season directional bias, with gains remaining stable across the five held-out outer test sets. Mechanism-group ablation and SHAP analyses showed that residual corrections were associated with stage-specific process-model timing, temperature structure, terrain/spatial, water-demand and radiation signals. Overall, this study improves the reliability of regional grapevine phenology prediction and provides a process-model and machine-learning residual-correction framework for vineyard management under complex terrain and heterogeneous climate conditions.

CRediT authorship contribution statement

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request, subject to the original data providers' terms.

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